

Composition vs. Volume

Political Alignment and Development Lending

May 2026

Main Findings

- ▶ **OLS:** Sector composition strongly predicts municipal GDP. Agriculture and Industry shares are positively associated with output; total loan volume is negligible.
- ▶ **IV:** The joint composition effect does not reject the null across most specifications. However, Industry shows individually significant positive coefficients (5%) in Governor and Mayor+Governor tiers. Total lending also shows no consistent effect.
- ▶ Findings hold across both classification schemes (broad sectors and firm-size groups) and weighting approaches.

Specification Overview

- ▶ **Outcome:** Log municipal GDP (municipality $\times$ year panel, 2002–2017).
- ▶ **Two sector classifications:**
 - ▶ Broad policy sectors: Agriculture, Industry, Infrastructure, Services.
 - ▶ Firm-size groups: Micro (≤ 9), Small (10–49), Medium (50–249), Large (≥ 250).
- ▶ **Four specifications:**
 - ▶ Pure OLS: use observed sector shares and total volume.
 - ▶ Composition IV: instrument sector shares only; control for total volume.
 - ▶ Full IV: instrument both sector shares and total volume jointly.
 - ▶ Volume IV: instrument total volume only; use observed sector shares.
- ▶ All specifications include municipality and year fixed effects.
- ▶ The highest mean-share category within each classification is the reference group.

Some Observations

- ▶ The weights used were selected after comparing multiple candidates.
- ▶ The production weight (owner-count intensity, full-municipality denominator) achieved the highest and most stable Kleibergen–Paap (KP) F-statistic across control specifications (≈ 60).
- ▶ Data concerns
 - ▶ The Agriculture sector has a lower firm-match rate with owner data (19%) than other sectors (67–95%), suggesting possible selection in this industry.
 - ▶ 14.6% of BNDES loan observations (2.5% by loan value) cannot be matched to RAIS employment data and fall outside the estimation sample.

Finalized Weights for the Instruments

- ▶ **Production weight (primary):** Weight each municipality–party cell by the number of politically affiliated firm owners, divided by the total owner count across *all* firms in the municipality.
- ▶ **Robustness weight (complement):** Weight by whether a firm has *any* aligned owner (a binary yes/no), divided by the total number of firms in the municipality.
- ▶ Both weights use the full local firm universe as the denominator — including firms that never received BNDES credit.
- ▶ The robustness weight captures the extensive margin of alignment (some affiliated owner). The production weight captures the intensive margin (more affiliated owners = stronger signal).

Instrument Strength and the AR Test

- ▶ **Total lending instrument** (Volume IV): the instrument for total BNDES volume achieves KP $F \approx 60$ (production) and ≈ 48 (robustness) under FE. Relevance is strong.
- ▶ **Sector-composition instruments** (Composition IV): per-variable first-stage F ranges 1.9–4.8.
- ▶ The composition Wald F reported on the result slides is the joint test of instrument relevance for the composition system.

Production Weight: Broad Policy Sectors

Weight: owner-count production weight.

Political tier	Pure OLS	Composition IV	Full IV	Volume IV
Mayor	11.86*** [<0.001]	0.07 [0.977]	0.06 [0.979]	1.78 [0.148]
Governor	11.86*** [<0.001]	2.59* [0.051]	1.58 [0.191]	1.00 [0.392]
Mayor + Governor	11.86*** [<0.001]	2.32* [0.073]	2.09* [0.099]	2.59* [0.051]
Mayor + President	11.86*** [<0.001]	0.29 [0.833]	0.08 [0.969]	1.78 [0.149]
Mayor + Gov. + President	11.86*** [<0.001]	1.76 [0.153]	1.55 [0.199]	2.53* [0.055]

Composition Wald F [p-value]. Controls: municipality and year fixed effects.

Production Weight: Broad Sectors — Pure OLS Coefficients

Weight: owner-count production weight.

Political tier	Agriculture	Industry	Infrastructure	Total
Mayor	0.028*** [<0.001]	0.016*** [<0.001]	-0.002 [0.649]	-2.34e-06 [0.975]
Governor	0.028*** [<0.001]	0.016*** [<0.001]	-0.002 [0.649]	-2.34e-06 [0.975]
Mayor + Governor	0.028*** [<0.001]	0.016*** [<0.001]	-0.002 [0.649]	-2.34e-06 [0.975]
Mayor + President	0.028*** [<0.001]	0.016*** [<0.001]	-0.002 [0.649]	-2.34e-06 [0.975]
Mayor + Gov. + President	0.028*** [<0.001]	0.016*** [<0.001]	-0.002 [0.649]	-2.34e-06 [0.975]

Entries are coefficient estimates with p-values in brackets. Omitted category: Services.

Production Weight: Broad Sectors — Composition IV Coefficients

Weight: owner-count production weight.

Political tier	Agriculture	Industry	Infrastructure	Total
Mayor	8.755 [0.689]	-4.291 [0.658]	5.235 [0.673]	0.002 [0.663]
Governor	0.427 [0.341]	0.532** [0.020]	-0.266 [0.487]	-2.48e-04 [0.136]
Mayor + Governor	0.308 [0.408]	0.394** [0.041]	-0.212 [0.432]	-1.81e-04 [0.212]
Mayor + President	0.068 [0.841]	0.056 [0.749]	-0.300 [0.352]	-2.95e-06 [0.984]
Mayor + Gov. + President	0.300 [0.282]	0.246* [0.068]	-0.201 [0.353]	-1.02e-04 [0.403]

Entries are coefficient estimates with p-values in brackets. Omitted category: Services.

Production Weight: Broad Sectors — Full IV Coefficients

Weight: owner-count production weight.

Political tier	Agriculture	Industry	Infrastructure	Total
Mayor	8.100 [0.673]	-3.309 [0.696]	4.634 [0.670]	0.030 [0.700]
Governor	-0.942 [0.493]	0.767** [0.037]	-0.235 [0.695]	0.028 [0.215]
Mayor + Governor	-0.740 [0.319]	0.681** [0.016]	-0.168 [0.754]	0.029* [0.050]
Mayor + President	0.076 [0.851]	0.097 [0.629]	-0.077 [0.867]	0.028* [0.075]
Mayor + Gov. + President	-0.087 [0.802]	0.327** [0.031]	-0.097 [0.735]	0.020* [0.063]

Entries are coefficient estimates with p-values in brackets. Omitted category: Services.

Production Weight: Broad Sectors — Volume IV Coefficients

Weight: owner-count production weight.

Political tier	Agriculture	Industry	Infrastructure	Total
Mayor	0.030 [0.111]	-0.098 [0.144]	-0.039 [0.278]	0.044 [0.165]
Governor	0.031 [0.247]	-0.152 [0.194]	-0.057 [0.344]	0.065 [0.164]
Mayor + Governor	0.030 [0.159]	-0.117** [0.023]	-0.045 [0.236]	0.052* [0.055]
Mayor + President	0.030 [0.111]	-0.098 [0.147]	-0.039 [0.280]	0.044 [0.167]
Mayor + Gov. + President	0.030 [0.159]	-0.117** [0.030]	-0.045 [0.244]	0.052* [0.065]

Entries are coefficient estimates with p-values in brackets. Omitted category: Services.

Production Weight: Firm-Size Groups

Weight: owner-count production weight.

Political tier	Pure OLS	Composition IV	Full IV	Volume IV
Mayor	34.84*** [<0.001]	0.09 [0.964]	0.12 [0.948]	0.85 [0.465]
Governor	34.84*** [<0.001]	2.24* [0.081]	1.62 [0.182]	0.32 [0.814]
Mayor + Governor	34.84*** [<0.001]	1.58 [0.193]	0.92 [0.428]	1.21 [0.303]
Mayor + President	34.84*** [<0.001]	1.19 [0.313]	0.59 [0.621]	0.82 [0.480]
Mayor + Gov. + President	34.84*** [<0.001]	1.79 [0.147]	0.82 [0.485]	1.21 [0.305]

Composition Wald F [p-value]. Controls: municipality and year fixed effects.

Production Weight: Firm-Size Groups — Pure OLS Coefficients

Weight: owner-count production weight.

Political tier	Small	Medium	Large	Total
Mayor	0.011*** [0.001]	0.040*** [<0.001]	0.052*** [<0.001]	-1.34e-05 [0.853]
Governor	0.011*** [0.001]	0.040*** [<0.001]	0.052*** [<0.001]	-1.34e-05 [0.853]
Mayor + Governor	0.011*** [0.001]	0.040*** [<0.001]	0.052*** [<0.001]	-1.34e-05 [0.853]
Mayor + President	0.011*** [0.001]	0.040*** [<0.001]	0.052*** [<0.001]	-1.34e-05 [0.853]
Mayor + Gov. + President	0.011*** [0.001]	0.040*** [<0.001]	0.052*** [<0.001]	-1.34e-05 [0.853]

Entries are coefficient estimates with p-values in brackets. Omitted category: Micro.

Production Weight: Firm-Size Groups — Composition IV Coefficients

Weight: owner-count production weight.

Political tier	Small	Medium	Large	Total
Mayor	1.681 [0.618]	3.116 [0.669]	-15.291 [0.666]	0.001 [0.707]
Governor	0.695** [0.047]	0.500 [0.639]	2.251 [0.388]	-3.94e-04 [0.112]
Mayor + Governor	0.618* [0.094]	-0.336 [0.640]	3.149 [0.187]	-2.67e-04 [0.439]
Mayor + President	0.328 [0.197]	-0.201 [0.557]	0.851 [0.288]	-2.08e-05 [0.865]
Mayor + Gov. + President	0.280 [0.250]	-0.324 [0.376]	1.575* [0.069]	-8.60e-05 [0.598]

Entries are coefficient estimates with p-values in brackets. Omitted category: Micro.

Production Weight: Firm-Size Groups — Full IV Coefficients

Weight: owner-count production weight.

Political tier	Small	Medium	Large	Total
Mayor	0.136 [0.959]	-1.233 [0.853]	7.376 [0.824]	0.044 [0.471]
Governor	0.567 [0.281]	0.472 [0.739]	2.624 [0.441]	0.029 [0.244]
Mayor + Governor	0.423 [0.410]	-0.481 [0.626]	4.267 [0.196]	0.035 [0.116]
Mayor + President	0.455 [0.189]	0.121 [0.785]	-0.116 [0.912]	0.040* [0.074]
Mayor + Gov. + President	0.241 [0.398]	0.002 [0.995]	0.947 [0.271]	0.027* [0.053]

Entries are coefficient estimates with p-values in brackets. Omitted category: Micro.

Production Weight: Firm-Size Groups — Volume IV Coefficients

Weight: owner-count production weight.

Political tier	Small	Medium	Large	Total
Mayor	0.003 [0.773]	-0.063 [0.183]	-0.049 [0.486]	0.043 [0.121]
Governor	-0.002 [0.896]	-0.125 [0.351]	-0.109 [0.488]	0.069 [0.209]
Mayor + Governor	0.002 [0.877]	-0.079* [0.066]	-0.065 [0.382]	0.050* [0.055]
Mayor + President	0.003 [0.730]	-0.058 [0.185]	-0.044 [0.486]	0.041 [0.102]
Mayor + Gov. + President	0.002 [0.873]	-0.079* [0.066]	-0.064 [0.381]	0.050* [0.054]

Entries are coefficient estimates with p-values in brackets. Omitted category: Micro.

Bottom Line

- ▶ **OLS tells a misleading story.** Sector composition correlates strongly with GDP in OLS because thriving municipalities attract specific types of lending — not because sectoral reallocation causes growth.
- ▶ **IV breaks the joint correlation.** Once composition is instrumented with political alignment shocks, the *joint* composition effect drops to zero across both classification schemes and both weighting approaches. Individual Industry coefficients remain significant at 5% in some tier specifications (Governor, Mayor+Governor under Composition IV and Full IV), a nuance worth investigating further.
- ▶ **Total lending is also not the driver.** Even when total BNDES volume is instrumented (Volume IV), there is no consistent positive effect on municipal GDP.
- ▶ **Implication.** Political alignment shapes *who receives* credit, but the sectoral and size composition of that credit does not independently predict local economic outcomes once the endogeneity is removed.

Technical Diagnostics

- ▶ Omitted reference category (owner-count production weight, Broad Policy Sectors): Services.
- ▶ Omitted reference category (owner-count production weight, Firm-Size Groups): Micro.
- ▶ Omitted reference category (aligned-firm robustness weight, Broad Policy Sectors): Services.
- ▶ Omitted reference category (aligned-firm robustness weight, Firm-Size Groups): Micro.
- ▶ Share of BNDES lending not classified by firm size: 20.9%.
- ▶ The instrument for total lending is constructed independently from the sector instruments (average correlation with row sum = 1.000).
- ▶ Agriculture sector firm-match rate: 19% (vs. 67–95% for other sectors) — possible selection in this industry.
- ▶ 14.6% of BNDES loan records (2.5% by loan value) have no RAIS employment match and fall outside the estimation sample.

Robustness Weight: Broad Policy Sectors

Weight: aligned-firm robustness weight.

Political tier	Pure OLS	Composition IV	Full IV	Volume IV
Mayor	11.86*** [<0.001]	0.65 [0.582]	0.63 [0.594]	3.46** [0.016]
Governor	11.86*** [<0.001]	1.40 [0.240]	0.58 [0.627]	0.32 [0.810]
Mayor + Governor	11.86*** [<0.001]	1.73 [0.159]	1.11 [0.344]	2.11* [0.096]
Mayor + President	11.86*** [<0.001]	1.98 [0.114]	1.47 [0.220]	2.32* [0.073]
Mayor + Gov. + President	11.86*** [<0.001]	1.62 [0.183]	1.48 [0.218]	2.10* [0.098]

Composition Wald F [p-value]. Controls: municipality and year fixed effects.

Robustness Weight: Broad Sectors — Pure OLS Coefficients

Weight: aligned-firm robustness weight.

Political tier	Agriculture	Industry	Infrastructure	Total
Mayor	0.028*** [<0.001]	0.016*** [<0.001]	-0.002 [0.649]	-2.34e-06 [0.975]
Governor	0.028*** [<0.001]	0.016*** [<0.001]	-0.002 [0.649]	-2.34e-06 [0.975]
Mayor + Governor	0.028*** [<0.001]	0.016*** [<0.001]	-0.002 [0.649]	-2.34e-06 [0.975]
Mayor + President	0.028*** [<0.001]	0.016*** [<0.001]	-0.002 [0.649]	-2.34e-06 [0.975]
Mayor + Gov. + President	0.028*** [<0.001]	0.016*** [<0.001]	-0.002 [0.649]	-2.34e-06 [0.975]

Entries are coefficient estimates with p-values in brackets. Omitted category: Services.

Robustness Weight: Broad Sectors — Composition IV Coefficients

Weight: aligned-firm robustness weight.

Political tier	Agriculture	Industry	Infrastructure	Total
Mayor	5.725 [0.412]	-0.230 [0.837]	0.288 [0.855]	3.61e-04 [0.543]
Governor	-0.165 [0.789]	0.640* [0.059]	-0.703 [0.154]	-3.04e-04 [0.206]
Mayor + Governor	0.137 [0.786]	0.413 [0.113]	-0.660* [0.062]	-1.70e-04 [0.402]
Mayor + President	1.094 [0.293]	0.483 [0.352]	-1.439** [0.021]	-1.17e-04 [0.741]
Mayor + Gov. + President	0.397 [0.338]	0.285 [0.165]	-0.443* [0.090]	-1.04e-04 [0.534]

Entries are coefficient estimates with p-values in brackets. Omitted category: Services.

Robustness Weight: Broad Sectors — Full IV Coefficients

Weight: aligned-firm robustness weight.

Political tier	Agriculture	Industry	Infrastructure	Total
Mayor	5.134 [0.444]	0.356 [0.835]	-0.313 [0.882]	0.021 [0.641]
Governor	-4.575 [0.342]	1.836 [0.210]	-0.965 [0.585]	0.096 [0.244]
Mayor + Governor	-1.690 [0.288]	0.920* [0.082]	-1.046 [0.161]	0.039 [0.111]
Mayor + President	1.187 [0.368]	0.876 [0.206]	-1.629** [0.036]	0.041 [0.131]
Mayor + Gov. + President	-0.006 [0.991]	0.397* [0.077]	-0.422 [0.126]	0.015 [0.210]

Entries are coefficient estimates with p-values in brackets. Omitted category: Services.

Robustness Weight: Broad Sectors — Volume IV Coefficients

Weight: aligned-firm robustness weight.

Political tier	Agriculture	Industry	Infrastructure	Total
Mayor	0.029*** [0.004]	-0.029 [0.531]	-0.016 [0.411]	0.017 [0.370]
Governor	0.035 [0.628]	-0.459 [0.360]	-0.157 [0.451]	0.184 [0.330]
Mayor + Governor	0.030 [0.192]	-0.131* [0.051]	-0.050 [0.276]	0.057* [0.086]
Mayor + President	0.029** [0.031]	-0.060 [0.242]	-0.026 [0.321]	0.029 [0.206]
Mayor + Gov. + President	0.030 [0.187]	-0.128* [0.055]	-0.049 [0.278]	0.056* [0.089]

Entries are coefficient estimates with p-values in brackets. Omitted category: Services.

Robustness Weight: Firm-Size Groups

Weight: aligned-firm robustness weight.

Political tier	Pure OLS	Composition IV	Full IV	Volume IV
Mayor	34.84*** [<0.001]	0.21 [0.889]	0.02 [0.997]	0.74 [0.526]
Governor	34.84*** [<0.001]	0.93 [0.427]	0.07 [0.974]	0.19 [0.902]
Mayor + Governor	34.84*** [<0.001]	1.68 [0.169]	1.15 [0.326]	1.04 [0.373]
Mayor + President	34.84*** [<0.001]	0.68 [0.564]	0.56 [0.644]	0.67 [0.573]
Mayor + Gov. + President	34.84*** [<0.001]	1.00 [0.394]	0.60 [0.617]	1.04 [0.373]

Composition Wald F [p-value]. Controls: municipality and year fixed effects.

Robustness Weight: Firm-Size Groups — Pure OLS Coefficients

Weight: aligned-firm robustness weight.

Political tier	Small	Medium	Large	Total
Mayor	0.011*** [0.001]	0.040*** [<0.001]	0.052*** [<0.001]	-1.34e-05 [0.853]
Governor	0.011*** [0.001]	0.040*** [<0.001]	0.052*** [<0.001]	-1.34e-05 [0.853]
Mayor + Governor	0.011*** [0.001]	0.040*** [<0.001]	0.052*** [<0.001]	-1.34e-05 [0.853]
Mayor + President	0.011*** [0.001]	0.040*** [<0.001]	0.052*** [<0.001]	-1.34e-05 [0.853]
Mayor + Gov. + President	0.011*** [0.001]	0.040*** [<0.001]	0.052*** [<0.001]	-1.34e-05 [0.853]

Entries are coefficient estimates with p-values in brackets. Omitted category: Micro.

Robustness Weight: Firm-Size Groups — Composition IV Coefficients

Weight: aligned-firm robustness weight.

Political tier	Small	Medium	Large	Total
Mayor	1.902 [0.454]	2.694 [0.518]	-11.674 [0.511]	9.38e-04 [0.637]
Governor	0.437 [0.266]	0.261 [0.740]	2.366 [0.233]	-3.59e-04 [0.235]
Mayor + Governor	0.522* [0.085]	-0.084 [0.854]	1.770 [0.255]	-1.65e-04 [0.457]
Mayor + President	0.420 [0.326]	0.644 [0.266]	-2.637 [0.197]	2.04e-04 [0.599]
Mayor + Gov. + President	0.300 [0.179]	-0.093 [0.740]	0.419 [0.654]	2.50e-06 [0.984]

Entries are coefficient estimates with p-values in brackets. Omitted category: Micro.

Robustness Weight: Firm-Size Groups — Full IV Coefficients

Weight: aligned-firm robustness weight.

Political tier	Small	Medium	Large	Total
Mayor	5.123 [0.842]	10.883 [0.861]	-49.031 [0.855]	-0.125 [0.904]
Governor	1.856 [0.861]	9.736 [0.731]	-10.760 [0.770]	0.475 [0.661]
Mayor + Governor	0.467 [0.267]	-0.150 [0.847]	2.585 [0.271]	0.035* [0.084]
Mayor + President	0.576 [0.229]	0.492 [0.527]	-2.047 [0.383]	0.040 [0.129]
Mayor + Gov. + President	0.358 [0.216]	0.189 [0.705]	0.134 [0.915]	0.035** [0.027]

Entries are coefficient estimates with p-values in brackets. Omitted category: Micro.

Robustness Weight: Firm-Size Groups — Volume IV Coefficients

Weight: aligned-firm robustness weight.

Political tier	Small	Medium	Large	Total
Mayor	0.007 [0.310]	-0.013 [0.740]	2.56e-04 [0.996]	0.022 [0.270]
Governor	-0.022 [0.591]	-0.387 [0.458]	-0.367 [0.502]	0.179 [0.366]
Mayor + Governor	8.98e-04 [0.935]	-0.088* [0.094]	-0.074 [0.394]	0.054* [0.076]
Mayor + President	0.006 [0.434]	-0.026 [0.520]	-0.012 [0.819]	0.028 [0.199]
Mayor + Gov. + President	8.93e-04 [0.936]	-0.089* [0.094]	-0.074 [0.394]	0.054* [0.076]

Entries are coefficient estimates with p-values in brackets. Omitted category: Micro.